

# Lecture Notes: Limits and Continuity

Calculus I

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## Section 1 One-Sided Limits

**Definition 1 (Left-Hand Limit)** Let  $f$  be a function defined on an open interval  $(c, a)$ . The left-hand limit of  $f(x)$  as  $x$  approaches  $a$  from the left is  $L$ , written

$$\lim_{x \rightarrow a^-} f(x) = L,$$

if for every  $\epsilon > 0$  there exists  $\delta > 0$  such that if  $a - \delta < x < a$ , then  $|f(x) - L| < \epsilon$ .

**Definition 2 (Right-Hand Limit)** Let  $f$  be a function defined on an open interval  $(a, c)$ . The right-hand limit of  $f(x)$  as  $x$  approaches  $a$  from the right is  $L$ , written

$$\lim_{x \rightarrow a^+} f(x) = L,$$

if for every  $\epsilon > 0$  there exists  $\delta > 0$  such that if  $a < x < a + \delta$ , then  $|f(x) - L| < \epsilon$ .

**Theorem 1 (Relationship Between One-Sided and Two-Sided Limits)** The two-sided limit  $\lim_{x \rightarrow a} f(x)$  exists and equals  $L$  if and only if both one-sided limits exist and are equal to  $L$ . That is,

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x) = L.$$

**Example 1** Consider the function  $f(x) = \frac{|x|}{x}$ . Find the one-sided limits as  $x \rightarrow 0$ . For  $x > 0$ ,  $|x| = x$ , so  $f(x) = 1$ . Thus,  $\lim_{x \rightarrow 0^+} f(x) = 1$ . For  $x < 0$ ,  $|x| = -x$ , so  $f(x) = -1$ . Thus,  $\lim_{x \rightarrow 0^-} f(x) = -1$ . Since the one-sided limits are different, the two-sided limit  $\lim_{x \rightarrow 0} \frac{|x|}{x}$  does not exist.

**Example 2** Consider the piecewise function  $g(x) = \begin{cases} x^2 + 1 & \text{if } x < 1 \\ 3x - 1 & \text{if } x \geq 1 \end{cases}$ . Evaluate the limits as  $x \rightarrow 1$ . Left-hand limit:  $\lim_{x \rightarrow 1^-} g(x) = \lim_{x \rightarrow 1^-} (x^2 + 1) = 1^2 + 1 = 2$ . Right-hand limit:  $\lim_{x \rightarrow 1^+} g(x) = \lim_{x \rightarrow 1^+} (3x - 1) = 3(1) - 1 = 2$ . Since both one-sided limits are equal to 2, the two-sided limit exists and  $\lim_{x \rightarrow 1} g(x) = 2$ . Note that  $g(1) = 3(1) - 1 = 2$ , so the function is continuous at  $x = 1$ .

## Section 2 Continuous Functions

**Definition 3 (Continuity at a Point)** A function  $f$  is continuous at a number  $a$  if the following three conditions are satisfied:

1.  $f(a)$  is defined (i.e.,  $a$  is in the domain of  $f$ ).
2.  $\lim_{x \rightarrow a} f(x)$  exists.
3.  $\lim_{x \rightarrow a} f(x) = f(a)$ .

If  $f$  is not continuous at  $a$ , we say  $f$  is discontinuous at  $a$ , or  $f$  has a discontinuity at  $a$ .

**Definition 4 (Continuity from the Left and Right)** A function  $f$  is continuous from the left at  $a$  if  $\lim_{x \rightarrow a^-} f(x) = f(a)$ . A function  $f$  is continuous from the right at  $a$  if  $\lim_{x \rightarrow a^+} f(x) = f(a)$ .

**Definition 5 (Continuity on an Interval)** A function  $f$  is continuous on an open interval  $(a, b)$  if it is continuous at every number in the interval. A function  $f$  is continuous on a closed interval  $[a, b]$  if it is continuous on  $(a, b)$ , continuous from the right at  $a$ , and continuous from the left at  $b$ .

**Theorem 2 (Properties of Continuous Functions)** If  $f$  and  $g$  are continuous at  $a$ , and  $c$  is a constant, then the following functions are also continuous at  $a$ :

1.  $f + g$
2.  $f - g$
3.  $cf$
4.  $fg$
5.  $\frac{f}{g}$ , provided  $g(a) \neq 0$ .

**Theorem 3 (Continuity of Composite Functions)** If  $f$  is continuous at  $b$  and  $\lim_{x \rightarrow a} g(x) = b$ , then  $\lim_{x \rightarrow a} f(g(x)) = f(b)$ . In particular, if  $g$  is continuous at  $a$  and  $f$  is continuous at  $g(a)$ , then the composite function  $f \circ g$  is continuous at  $a$ .

**Theorem 4 (Continuity of Elementary Functions)** The following types of functions are continuous at every number in their domains:

- Polynomials
- Rational functions
- Root functions
- Trigonometric functions
- Exponential functions
- Logarithmic functions

**Example 3** Evaluate  $\lim_{x \rightarrow \pi} \frac{\sin x}{x^2 + 1}$ . Since  $\sin x$  and  $x^2 + 1$  are continuous everywhere, and  $x^2 + 1 \neq 0$  for all  $x$ , the function  $f(x) = \frac{\sin x}{x^2 + 1}$  is continuous at  $x = \pi$ . Therefore, we can simply substitute:

$$\lim_{x \rightarrow \pi} \frac{\sin x}{x^2 + 1} = \frac{\sin \pi}{\pi^2 + 1} = \frac{0}{\pi^2 + 1} = 0.$$

## Section 3 Types of Discontinuities

**Definition 6 (Removable Discontinuity)** A discontinuity at  $a$  is called removable if  $\lim_{x \rightarrow a} f(x)$  exists but is not equal to  $f(a)$  (or  $f(a)$  is undefined). The graph has a hole at  $a$ . The discontinuity can be "removed" by redefining  $f(a)$  to be the limit value.

**Example 4** The function  $f(x) = \frac{x^2 - 4}{x - 2}$  has a removable discontinuity at  $x = 2$ . Although  $f(2)$  is undefined,  $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = 4$ .

**Definition 7 (Jump Discontinuity)** A discontinuity at  $a$  is called a jump discontinuity if the one-sided limits  $\lim_{x \rightarrow a^-} f(x)$  and  $\lim_{x \rightarrow a^+} f(x)$  both exist but are not equal. The graph "jumps" from one value to another.

**Example 5** The function  $f(x) = \begin{cases} x & \text{if } x < 0 \\ x + 1 & \text{if } x \geq 0 \end{cases}$  has a jump discontinuity at  $x = 0$ . The left limit is 0 and the right limit is 1.

**Definition 8 (Infinite Discontinuity)** A discontinuity at  $a$  is called an infinite discontinuity if at least one of the one-sided limits is  $\infty$  or  $-\infty$ . This typically corresponds to a vertical asymptote.

**Example 6** The function  $f(x) = \frac{1}{x^2}$  has an infinite discontinuity at  $x = 0$ . As  $x \rightarrow 0$ ,  $f(x) \rightarrow \infty$ .

**Definition 9 (Oscillating Discontinuity)** A discontinuity at  $a$  is called an oscillating discontinuity if the function oscillates infinitely often between two values as  $x$  approaches  $a$ , so the limit does not exist.

**Example 7** The function  $f(x) = \sin\left(\frac{1}{x}\right)$  has an oscillating discontinuity at  $x = 0$ . As  $x \rightarrow 0$ , the function oscillates between  $-1$  and  $1$  infinitely many times.

## Section 4 The Intermediate Value Theorem

**Theorem 5 (Intermediate Value Theorem (IVT))** Suppose that  $f$  is continuous on the closed interval  $[a, b]$  and let  $N$  be any number between  $f(a)$  and  $f(b)$ , where  $f(a) \neq f(b)$ . Then there exists a number  $c$  in  $(a, b)$  such that  $f(c) = N$ .

We prove the case where  $f(a) < N < f(b)$ . Let  $S = \{x \in [a, b] \mid f(x) < N\}$ . Since  $a \in S$ ,  $S$  is non-empty, and since  $S \subseteq [a, b]$ , it is bounded above. By the Completeness Axiom of real numbers,  $S$  has a least upper bound. Let  $c = \sup S$ . We claim  $f(c) = N$ . Suppose  $f(c) < N$ . Since  $f$  is continuous at  $c$ , there exists  $\delta > 0$  such that for all  $x \in (c - \delta, c + \delta) \cap [a, b]$ ,  $f(x) < N$ . This implies there are numbers greater than  $c$  in  $S$ , contradicting that  $c$  is an upper bound. Suppose  $f(c) > N$ . Again, by continuity, there exists  $\delta > 0$  such that  $f(x) > N$  for all  $x \in (c - \delta, c + \delta) \cap [a, b]$ . This implies  $c - \delta$  is an upper bound for  $S$ , contradicting that  $c$  is the least upper bound. Therefore,  $f(c) = N$ .

**Example 8 (Root Finding)** Show that the equation  $x^3 - x - 1 = 0$  has a root in the interval  $(1, 2)$ . Let  $f(x) = x^3 - x - 1$ . Since  $f$  is a polynomial, it is continuous everywhere.  $f(1) = 1^3 - 1 - 1 = -1 < 0$ .  $f(2) = 2^3 - 2 - 1 = 8 - 2 - 1 = 5 > 0$ . Since  $f(1) < 0 < f(2)$ , by the Intermediate Value Theorem, there exists at least one number  $c \in (1, 2)$  such that  $f(c) = 0$ . Thus, the equation has a root in  $(1, 2)$ .

**Example 9 (Fixed Point)** Show that the equation  $\cos x = x$  has a solution. Let  $f(x) = \cos x - x$ . We seek  $x$  such that  $f(x) = 0$ .  $f(0) = \cos 0 - 0 = 1 > 0$ .  $f\left(\frac{\pi}{2}\right) = \cos \frac{\pi}{2} - \frac{\pi}{2} = 0 - \frac{\pi}{2} = -\frac{\pi}{2} < 0$ . Since  $f$  is continuous and  $f(0) > 0 > f\left(\frac{\pi}{2}\right)$ , by the IVT, there exists  $c \in (0, \frac{\pi}{2})$  such that  $f(c) = 0$ , i.e.,  $\cos c = c$ .

## Section 5 Limits Involving Infinity

**Definition 10 (Limits at Infinity)** Let  $f$  be a function defined on  $(a, \infty)$ . Then

$$\lim_{x \rightarrow \infty} f(x) = L$$

means that for every  $\epsilon > 0$  there exists a number  $N$  such that if  $x > N$ , then  $|f(x) - L| < \epsilon$ . Similarly, for  $f$  defined on  $(-\infty, a)$ ,

$$\lim_{x \rightarrow -\infty} f(x) = L$$

means that for every  $\epsilon > 0$  there exists a number  $N$  such that if  $x < N$ , then  $|f(x) - L| < \epsilon$ .

**Definition 11 (Horizontal Asymptote)** The line  $y = L$  is called a horizontal asymptote of the curve  $y = f(x)$  if either  $\lim_{x \rightarrow \infty} f(x) = L$  or  $\lim_{x \rightarrow -\infty} f(x) = L$ .

**Theorem 6 (Limits at Infinity of Basic Functions)** For any rational number  $r > 0$ :

$$\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0 \quad \text{and} \quad \lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0 \quad (\text{if } x^r \text{ is defined for } x < 0).$$

**Example 10** Evaluate  $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x + 1}{5x^2 + 4x - 7}$ . Divide numerator and denominator by the highest power of  $x$  in the denominator, which is  $x^2$ :

$$\lim_{x \rightarrow \infty} \frac{3 - \frac{2}{x} + \frac{1}{x^2}}{5 + \frac{4}{x} - \frac{7}{x^2}} = \frac{3 - 0 + 0}{5 + 0 - 0} = \frac{3}{5}.$$

Thus, the line  $y = \frac{3}{5}$  is a horizontal asymptote.

**Definition 12 (Infinite Limits)** Let  $f$  be a function defined on an open interval containing  $a$ , except possibly at  $a$ . Then

$$\lim_{x \rightarrow a} f(x) = \infty$$

means that for every  $M > 0$  there exists  $\delta > 0$  such that if  $0 < |x - a| < \delta$ , then  $f(x) > M$ . Similarly,  $\lim_{x \rightarrow a} f(x) = -\infty$  means that for every  $M > 0$  there exists  $\delta > 0$  such that if  $0 < |x - a| < \delta$ , then  $f(x) < -M$ .

**Definition 13 (Vertical Asymptote)** The line  $x = a$  is called a vertical asymptote of the curve  $y = f(x)$  if at least one of the following holds:  $\lim_{x \rightarrow a^-} f(x) = \pm\infty$ ,  $\lim_{x \rightarrow a^+} f(x) = \pm\infty$ , or  $\lim_{x \rightarrow a} f(x) = \pm\infty$ .

**Example 11** Find the vertical asymptotes of  $f(x) = \frac{x+2}{x^2-4}$ . Factor the denominator:  $x^2 - 4 = (x-2)(x+2)$ . Thus,  $f(x) = \frac{x+2}{(x-2)(x+2)}$ . For  $x \neq -2$ ,  $f(x) = \frac{1}{x-2}$ . As  $x \rightarrow 2^+$ ,  $f(x) \rightarrow \infty$ . As  $x \rightarrow 2^-$ ,  $f(x) \rightarrow -\infty$ . Therefore,  $x = 2$  is a vertical asymptote. At  $x = -2$ , there is a removable discontinuity (a hole), not a vertical asymptote.

**Theorem 7 (End Behavior of Polynomials)** For a polynomial  $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$ , the end behavior is determined by the leading term  $a_n x^n$ . That is,

$$\lim_{x \rightarrow \infty} P(x) = \lim_{x \rightarrow \infty} a_n x^n \quad \text{and} \quad \lim_{x \rightarrow -\infty} P(x) = \lim_{x \rightarrow -\infty} a_n x^n.$$

**Theorem 8 (End Behavior of Rational Functions)** For a rational function  $f(x) = \frac{P(x)}{Q(x)}$ , where  $P(x)$  and  $Q(x)$  are polynomials with leading terms  $a_n x^n$  and  $b_m x^m$  respectively:

1. If  $n < m$ , then  $\lim_{x \rightarrow \pm\infty} f(x) = 0$  (horizontal asymptote  $y = 0$ ).
2. If  $n = m$ , then  $\lim_{x \rightarrow \pm\infty} f(x) = \frac{a_n}{b_m}$  (horizontal asymptote  $y = \frac{a_n}{b_m}$ ).
3. If  $n > m$ , then  $\lim_{x \rightarrow \pm\infty} f(x) = \pm\infty$  (no horizontal asymptote; the limit is infinite).

## Section 6 Relative Rates of Growth

**Definition 14 (Faster, Slower, Same Rate)** Let  $f$  and  $g$  be positive functions for sufficiently large  $x$ .

- $f$  grows faster than  $g$  as  $x \rightarrow \infty$  if  $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \infty$ . We write  $f \gg g$ .
- $f$  grows slower than  $g$  as  $x \rightarrow \infty$  if  $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0$ . We write  $f \ll g$ .
- $f$  and  $g$  grow at the same rate as  $x \rightarrow \infty$  if  $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L$ , where  $L$  is a positive finite constant.

**Definition 15 (Big-O and Little-o Notation)** Let  $f$  and  $g$  be functions defined for sufficiently large  $x$ .

- $f(x) = O(g(x))$  as  $x \rightarrow \infty$  if there exist constants  $M > 0$  and  $x_0$  such that  $|f(x)| \leq M|g(x)|$  for all  $x > x_0$ .
- $f(x) = o(g(x))$  as  $x \rightarrow \infty$  if  $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0$ .

**Theorem 9 (Hierarchy of Growth Rates)** As  $x \rightarrow \infty$ , the following hierarchy holds for positive constants  $a > 0$ ,  $b > 1$ , and  $p > 0$ :

$$\ln x \ll x^p \ll b^x \ll x^x.$$

More generally, any logarithmic function grows slower than any positive power function, which grows slower than any exponential function with base greater than 1.

**Example 12** Compare the growth rates of  $f(x) = x^2$  and  $g(x) = e^x$  as  $x \rightarrow \infty$ .

$$\lim_{x \rightarrow \infty} \frac{x^2}{e^x} = 0.$$

Therefore,  $x^2 \ll e^x$ . The exponential function grows much faster than the polynomial.

**Example 13** Compare the growth rates of  $f(x) = \ln x$  and  $g(x) = \sqrt{x}$  as  $x \rightarrow \infty$ .

$$\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}} = 0.$$

Therefore,  $\ln x \ll \sqrt{x}$ . The square root function grows faster than the natural logarithm.

### 6.1 Table of Equivalent Functions (Asymptotic Equivalences)

**Definition 16 (Asymptotic Equivalence)** Two functions  $f(x)$  and  $g(x)$  are asymptotically equivalent as  $x \rightarrow a$  (where  $a$  can be a finite number or  $\pm\infty$ ), denoted  $f(x) \sim g(x)$ , if

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 1.$$

The following table provides a list of common asymptotic equivalences used in calculus, particularly useful for evaluating limits and analyzing relative rates of growth.

| Function                          | Asymptotic Equivalent | Limit Condition           |
|-----------------------------------|-----------------------|---------------------------|
| $\sin x$                          | $x$                   | $x \rightarrow 0$         |
| $\tan x$                          | $x$                   | $x \rightarrow 0$         |
| $\arcsin x$                       | $x$                   | $x \rightarrow 0$         |
| $\arctan x$                       | $x$                   | $x \rightarrow 0$         |
| $1 - \cos x$                      | $\frac{x^2}{2}$       | $x \rightarrow 0$         |
| $e^x - 1$                         | $x$                   | $x \rightarrow 0$         |
| $\ln(1 + x)$                      | $x$                   | $x \rightarrow 0$         |
| $(1 + x)^\alpha - 1$              | $\alpha x$            | $x \rightarrow 0$         |
| $\ln x$                           | $x - 1$               | $x \rightarrow 1$         |
| $a^x - 1$                         | $x \ln a$             | $x \rightarrow 0$         |
| $P_n(x) = a_n x^n + \cdots + a_0$ | $a_n x^n$             | $x \rightarrow \pm\infty$ |

**Example 14** Evaluate  $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$ . Using the asymptotic equivalence from the table,  $1 - \cos x \sim \frac{x^2}{2}$  as  $x \rightarrow 0$ . Therefore,

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{\frac{x^2}{2}}{x^2} = \frac{1}{2}.$$

**Example 15** Evaluate  $\lim_{x \rightarrow 0} \frac{\ln(1 + x)}{x}$ . Using the asymptotic equivalence,  $\ln(1 + x) \sim x$  as  $x \rightarrow 0$ . Therefore,

$$\lim_{x \rightarrow 0} \frac{\ln(1 + x)}{x} = \lim_{x \rightarrow 0} \frac{x}{x} = 1.$$

**Example 16** Evaluate  $\lim_{x \rightarrow \infty} \frac{x^3 + 2x^2 + 1}{4x^3 - x + 5}$ . Using the asymptotic equivalence for polynomials as  $x \rightarrow \infty$ ,  $x^3 + 2x^2 + 1 \sim x^3$  and  $4x^3 - x + 5 \sim 4x^3$ . Therefore,

$$\lim_{x \rightarrow \infty} \frac{x^3 + 2x^2 + 1}{4x^3 - x + 5} = \lim_{x \rightarrow \infty} \frac{x^3}{4x^3} = \frac{1}{4}.$$

**Theorem 10 (Growth Rate Comparisons Involving Logarithms and Exponentials)** For any constant  $p > 0$  and any base  $b > 1$ :

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\ln x}{x^p} &= 0, \\ \lim_{x \rightarrow \infty} \frac{x^p}{b^x} &= 0, \\ \lim_{x \rightarrow \infty} \frac{b^x}{x^p} &= 0. \end{aligned}$$

These limits formalize the hierarchy: logarithmic growth is slower than polynomial growth, which is slower than exponential growth, which is slower than factorial or  $x^x$  growth.

**Example 17** Determine which function grows faster as  $x \rightarrow \infty$ :  $f(x) = x^{100}$  or  $g(x) = 2^x$ . Despite the large exponent on  $x$ , the exponential function eventually outpaces any polynomial. By the theorem above with  $p = 100$  and  $b = 2$ :

$$\lim_{x \rightarrow \infty} \frac{x^{100}}{2^x} = 0.$$

Therefore,  $x^{100} \ll 2^x$ .